2026

Great Lakes Water Quality

COMP-3250A

FEBURARY, 10TH 2026

GROUP MEMBERS (GROUP 4)

First & Last Name	Student ID	Email
Aaron Sinn	Sinna (110071649)	sinna@uwindsor.ca
Muhammad Bagalagel	110113304	bagalag1@uwindsor.ca
Daniel Laurin	110078719	Laurind@uwindsor.ca
Marko Koljivrat	110112564	Koljivr1@uwindsor.ca

CONTENTS

Data Analytics on Great Lakes Water Quality 2
About the Project 2
Abstract 2
Problem Description 2
Motivation for the Problem 3
Solution Statement 3
Technologies Used 4
Team Member Roles 4
Methodology 5
Proposed Project Timeline, Deliverables 6
References 7

DATA ANALYTICS ON GREAT LAKES WATER QUALITY

ABOUT THE PROJECT

Project Title & Description	• Analyse and visualize water quality data from the Great Lakes, while using DataStream an aggregator of Community-Based Water Monitoring (CBWM) and research institutes. And apply machine learning to interpolate water quality values (Chloride, Phosphorus, Ph, E. coli etc.) in geographic areas where data is absent
Key Objectives & Desired Outcomes	• **Retrieve** and use Pandas to normalize the WQX schema • **Implement** K-Nearest Neighbours (KNN) regression algorithm using Scikit-Learn to fill gaps in the data • **Develop** a dynamic Tableau dashboard and have interactive filter mechanism • **Identify** and document specific patterns in water contamination that would impact the Windsor-Esser region • **Respect** and handle all the data with OCAP principle
Team Members	• Aaron Sinn – 4th computer science student intrested in data analytics • Muhammad Bagalagel - 4th year Supply Chain and Data Analytics with Computer Science student • Daniel Laurin – 4th year BIT honours, studied PMP, CCNA, Sec+ certificates • Marko Koljivrat – 4th year Business and Computer Science Student specializing in Supply Chain & Business Analytics

ABSTRACT

The goal of our project is to analyze and visualize water quality data from the Great Lakes using open data sets provided by DataStream. We plan to focus water qualitative on sample data collected between 2021 - 2025 and examine specific characteristics such as Chloride, Phosphorus, Ph, E. coli and other indicators that reflect overall water quality. Through this project, we want to explore noticeable patterns, trends, or unusual values across different regions of the Great Lakes.

We will achieve this by building a data pipeline using Python (Pandas) to query the DataStream API and extract specific subsets of data relevant to given time frame. We will then clean and normalize these data points to handle the complex Water Quality Exchange (WQX) xml schema. To address the gaps in the data, we will use Scikit-Learn to build a K-Nearest Neighbors (KNN) model that predicts water quality trends on unmonitored coordinates.

The final result of our project will be an interactive dashboard that allows users to view water quality indicators by location and time. Users will be able to filter the data and observe trends across different areas of the Great Lakes. Along with the dashboard, we will provide a written explanation of our analysis process and the insights we discovered. Overall, our project aims to show how open environmental data can be transformed into meaningful visual insights using data analytics techniques learned in this course.

PROBLEM DESCRIPTION

As tech companies begin to settle down and build infrastructure for large AI data centers near the great lakes, water levels have gone down, and risk of pollution has begun to increase. AI data centers being situated around the great lakes were inevitable as AI is the largest technological innovation of the 21st century and requirements for fresh water to cool these high-usage, high-velocity generative AI models grow with use. AI is being pushed globally, and little has been regulated to protect those around the infrastructure.

As mentioned in this proposal citizens around these areas rely on this water for all their needs (drinking, bathing, cooking, etc.) and to have dynamic and extensive data and visual tools about the quality of this water and any pollutants are beyond valuable. Instead of focusing on one pollutant, we want to inform end users about the entire diagnostics and provide a sense of security to those around the great lakes. Larger data centers are newer to the area or are only a proposal currently, water is ever flowing, and pollutants can be introduced into new bodies of water every day coming from anywhere in North America.

Water quality is not something that is on the everyday citizens until the quality of your home water is noticeably poor. We want to create an easy use dashboard and map, mapping the water in an area and quality and pollutant diagnostics that is accessible for everyone. Water is in demand more than ever, and we want to make sure that it stays safe for the people.

MOTIVATION FOR THE PROBLEM

Our motivation to take up this problem is both environmental concern and data literacy. The Great Lakes are a freshwater source for millions of people and are of extreme significance in Windsor-Essex County. As students of data analytics, we are driven by the opportunity to apply our technical skills to datasets that have tangible impacts on our daily lives.

This problem is interesting because the Great Lakes represent a massive ecosystem that's directly tied to much of Canada and Windsor-Essex County. By focusing on the 2021-2025 window, we will be able to see how specific indicators such chlorine affects water quality in recent years and what trends may continue into the future. This dataset will provide us with the opportunity to show how raw and complex data can be made accessible in an intuitive way.

This problem is challenging because of the sheer size of the dataset. Data spanning multiple years will require and numerous characteristics will require sophisticated filtering to avoid irrelevant data. Additionally, the Great Lakes water quality dataset may contain missing values which will require careful data cleansing to ensure reliability and accuracy.

People who rely on the Great Lakes for drinking water and recreation will benefit from a solution to this problem. By transforming raw data into an intuitive visual format, these residents will be able to have a better understanding of the water quality that's in their "backyard". Policy makers and urban planners can utilize these data insights to justify investments or environmental regulations regarding the Great Lakes.

SOLUTION STATEMENT

TECHNOLOGIES USED

Python (Pandas & Numpy): Selected as our primary tool for ETL (Extract, Transform, Load). Its library ecosystem is designed for the manipulation of tabular data, programmatically handles null values and filters out noisy data points from the DataStream API.

Scikit-Learn (Machine Learning): Selected to implement K-Nearest Neighbors (KNN) Regression. This will allow us to mathematically predict water quality in the gaps between monitoring collections.

Tableau: Selected because it is a proven industry standard for visualization. It provides the ability to create intuitive maps and drill-down dashboard. It is cross-platform and allows for the integration of our Python-generated dataset. Tableau is also cross platform and runs on Windows, MacOS, and Linux.

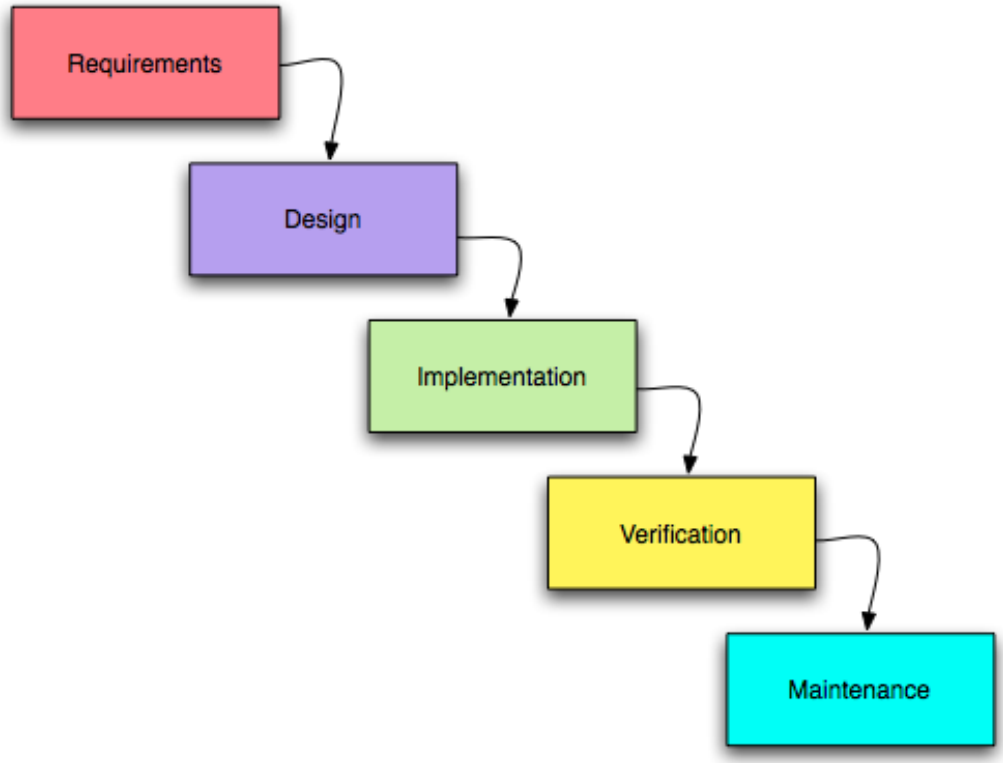
Teams/ Discord: For collaboration and team management we choose two platforms. Teams the ability to track our workflow and upcoming due dates. Discord for direct communication and file sharing.

TEAM MEMBER ROLES

Role Title	Role Objectives
Data Engineer (Aaron)	Focuses on merging data from the DataStream API and writing Python scripts to cleanse and process the WQX schema.
Data Analyst (Muhammad)	Focuses on exploratory analysis to identify patterns/trends and decides which chemical features (e.g., Chloride, pH) are significant for the model
UI/UX Architect (Marko)	Focuses on building the interactive dashboard and map layers using the processed data and ML predictions
Project Manager (Daniel)	Manages the Teams workspace, ensures Waterfall adherence, and acts as the primary point of contact with Meghan McLeod, DataStream Data-Specialist

METHODOLOGY

We chose **Waterfall Methodology** (PMBOK). The requirements for this project (fixed deadline, defined API, specific deliverables) are fixed and will not be constantly changing. Waterfall allows us to map out exactly how much time is needed for each stage (Ingestion, Cleaning, Modeling, and Visualization), ensuring we meet the March 29th deadline.



WHY DID WE CHOOSE IT?

We chose Great Lakes DataStream comprehensive coverage, allowing us to analyze recent industrial impacts like AI data centers. Additionally, team member Daniel has direct contact with developer access to facilitate API integration

IS THE SOURCE CREDIBLE?

Yes. It aggregates data from Environment and Climate Change Canada (ECCC). It is an initiative of The Gordon Foundation (partnered with RBC) and follows strict US, EPA, WQX standards.

Features of the Dataset

- ***MonitoringLocationIdentifier:*** coordinates (Lat/Long)
- ***AcitvityStartDate:*** timestamp for time analysis (2021-2025)
- ***CharacterticName:*** specific pollutant (e.g., “Chloride”)
- ***ResultMeasureValue:*** the numeric value used for our ML mode

PROPOSED PROJECT TIMELINE, DELIVERABLES

ESTIMATED PROJECT COMPLETION DATE: 29/03/26

Phase 1: [13/02/26 – 23/02/26]

Key Deliverable	Responsibility to Key Objectives	Due Date
Project Proposal	finalize project charter and technologies	13/02/26
Onboarding	Request API key to Meghan McLeod at DataStream and do meeting advancing our proposal	23/02/26

Phase 2: [5/03/26 – 22/03/26]

Key Deliverable	Responsibility to Key Objectives	Due Date
Data Engineering	Scripting the API pull and cleaning data	5/03/26
Machine Learning	Training the KNN model	15/03/26
Visualization	Building the Tableau Dashboard	22/03/26

Phase 3: [22/03/26– 29/03/26]

Key Deliverable	Responsibility to Key Objectives	Due Date
Reporting and Presentation prep	Final report writing and presentation preparation	29/03/26

REFERENCES

PRIMARY SOURCES

Canada, Environment and Climate Change. "Government of Canada." *Canada.Ca*, / Gouvernement du Canada, 9 July 2024, www.canada.ca/en/environment-climate-change/services/water-overview/publications/canada-water-act-2022-2023.html.

"Canada's Open Access Platform for Sharing Water Data." *DataStream*, datastream.org/en-ca/platform. Accessed 13 Feb. 2026.

Guerin, Andrew J., et al. "High-Resolution Freshwater Dissolved Calcium and Ph Data Layers for Canada and the United States." *Nature News*, Nature Publishing Group, 11 Apr. 2024, www.nature.com/articles/s41597-024-03165-8.

"Meaningful Water Policy and Governance Reform in Canada | Department of Economic and Social Affairs." *United Nations*, United Nations, sdgs.un.org/partnerships/meaningful-water-policy-and-governance-reform-canada. Accessed 13 Feb. 2026.

"Meaningful Water Policy and Governance Reform in Canada | Department of Economic and Social Affairs." *United Nations*, United Nations, sdgs.un.org/partnerships/meaningful-water-policy-and-governance-reform-canada. Accessed 13 Feb. 2026.

TECHNOLOGIES

"1.6. Nearest Neighbors." *Scikit*, scikit-learn.org/stable/modules/neighbors.html. Accessed 13 Feb. 2026.

"Pandas." *Pandas*, pandas.pydata.org/. Accessed 13 Feb. 2026.

"Tableau for Students." *Tableau*, www.tableau.com/academic/students. Accessed 13 Feb. 2026.

"Welcome to Python.Org." *Python.Org*, www.python.org/. Accessed 13 Feb. 2026.

WBS AND PROJECT MANAGEMENT

Microsoft Planner, planner.cloud.microsoft/webui/. Accessed 13 Feb. 2026.

PMBOK GUIDE | Project Management Institute, www.pmi.org/standards/pmbok. Accessed 14 Feb. 2026.

"What Is the Waterfall Methodology?" *Atlassian*, Atlassian, 6 June 2025, www.atlassian.com/agile/project-management/waterfall-methodology.

"What Is the Waterfall Methodology?" *Atlassian*, Atlassian, 6 June 2025, www.atlassian.com/agile/project-management/waterfall-methodology.